Lab: Azure AI Foundry Chat Playground

Lab Overview

In this lab, you will use Microsoft Azure and Azure AI Foundry to deploy and interact with a Large Language Model (LLM) in the Chat Playground. You will verify the gpt-4o model, configure system instructions, test prompts, and adjust parameters such as temperature to observe output variations.

This lab focuses on understanding how instructions and parameter tuning influence AI responses in real time.

By the end of this lab, you will:

- Access an Azure AI Hub and project
- Verify or deploy the gpt-4o model
- Configure system instructions
- Test prompts and review responses
- Tune parameters to compare output behaviour

What You Will Learn

After completing this lab, you will be able to:

- Navigate Azure AI Foundry and open Chat Playground
- Deploy or verify a base LLM model
- Configure system instructions to guide responses
- Test prompts and evaluate outputs
- Adjust temperature settings
- Compare structured vs. creative responses

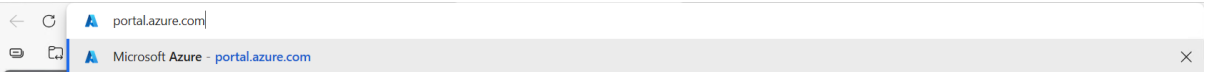
Prerequisites

Before starting, ensure you have:

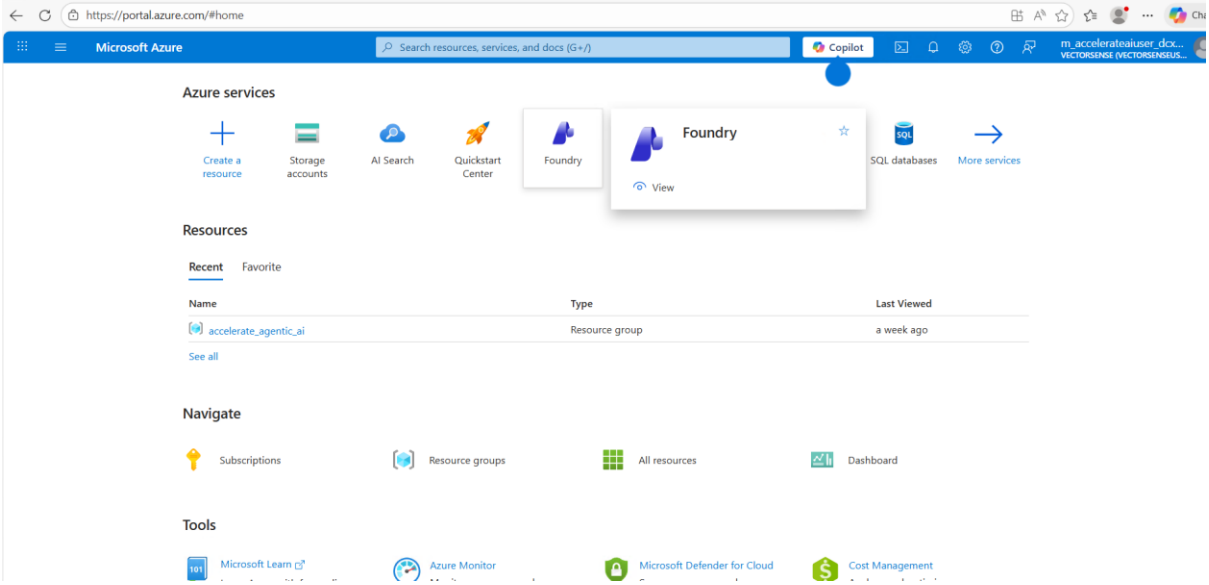
- Valid Azure Portal lab credentials
- Access to an Azure AI Hub and project
- A deployed gpt-4o model (or ability to deploy one)
- A supported browser (Microsoft Edge recommended)

Step by step Instructions

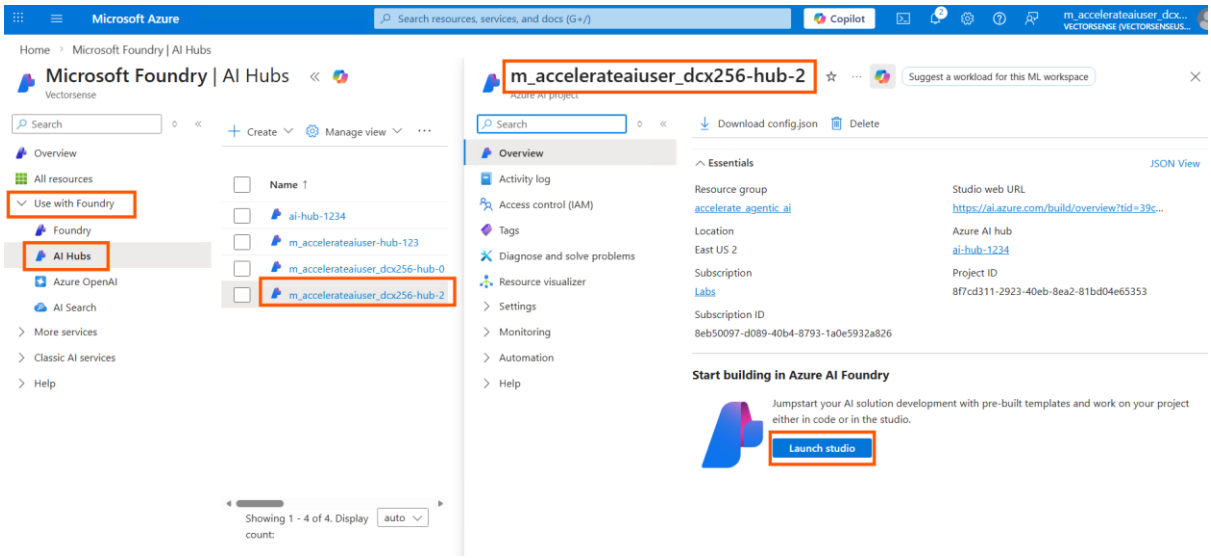
1. Open **Microsoft Edge** and go to https://portal.azure.com. Sign in with the provided credentials.



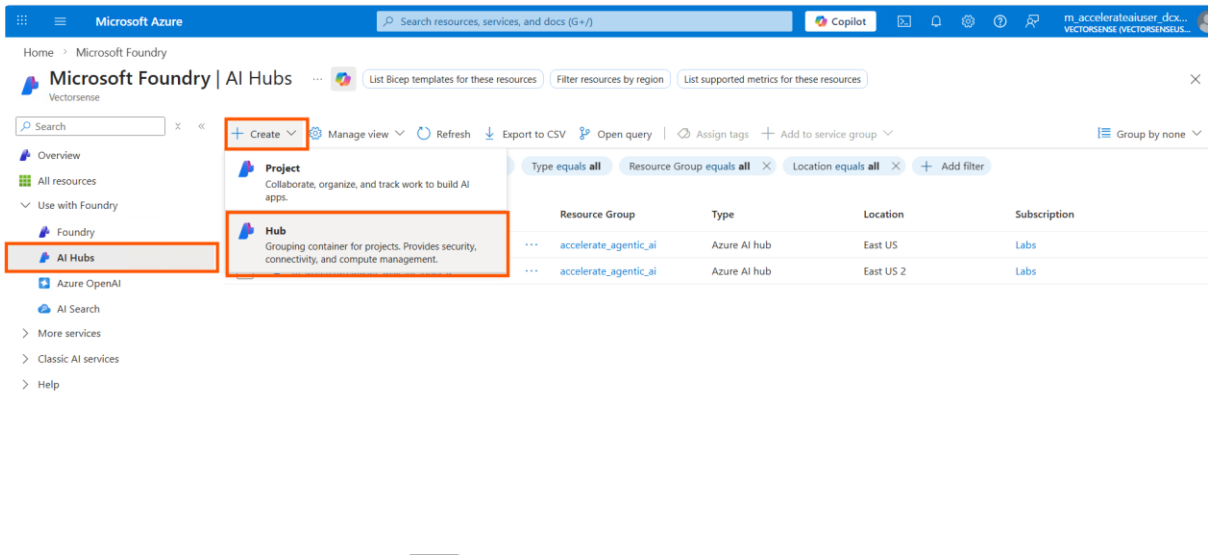
2. Search **Foundry** or click on **Foundry** from the main menu.



3. Click on **AI Hubs** under **Use with Foundry** and click on **m_accelerateaiuser_dcx256-hub-2** (**Azure AI project)** as shown in the image. Click on **Launch studio.**
   <u>We have already deployed the Azure AI Hub, project and model so you can directly **jump to step 10**.</u> The below instructions can be used if you cannot find the deployed hub and project.

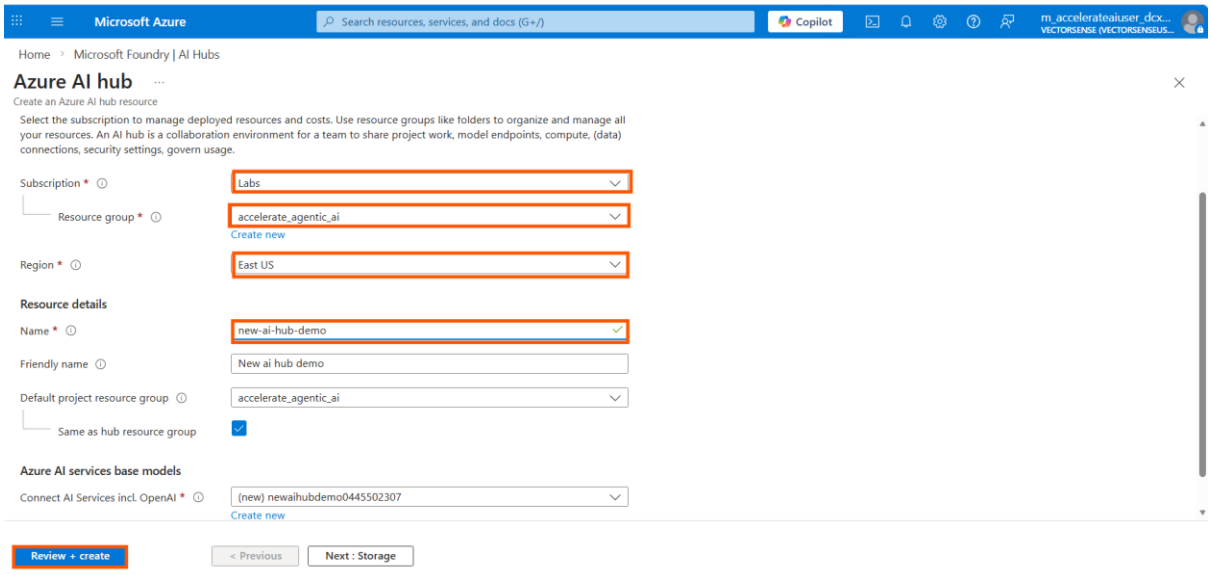


4. Under Microsoft Foundry, click on **AI Hubs -> Create -> Hub.**

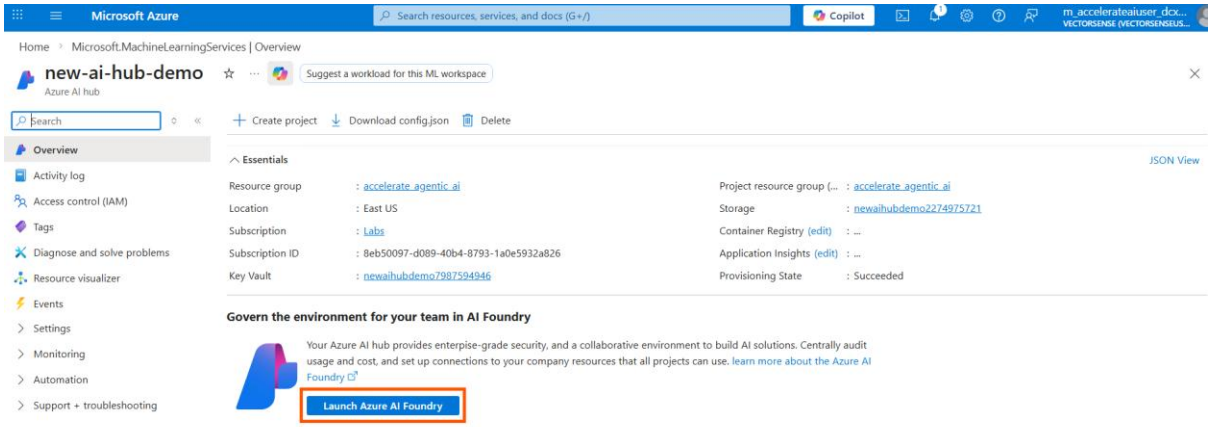


5. Under **Azure AI Hub,** configure. Click on **Review + create** -> **Create**:

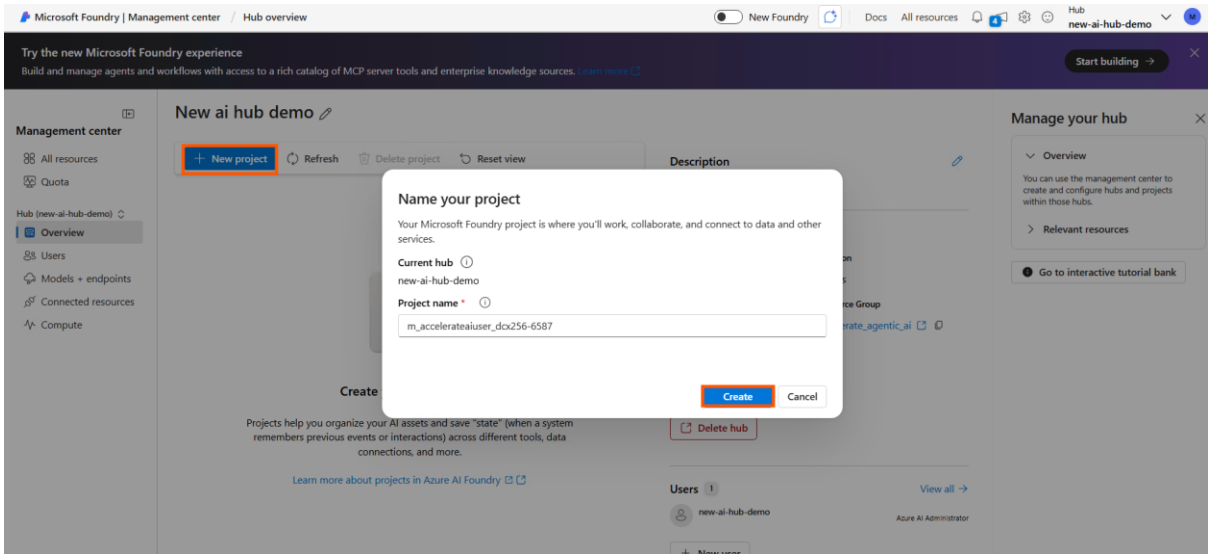
- Subscription: Labs
- Resource Group: Create new or default
- Region: East US/France Central/Sweden Central
- Name: Unique name



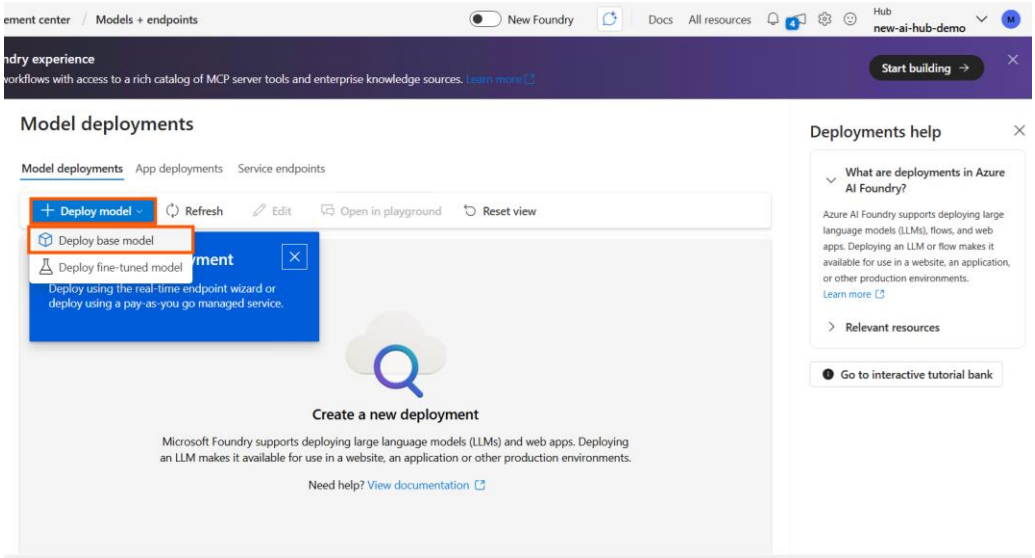
6. Click on **Go to resource -> Launch Azure AI Foundry**



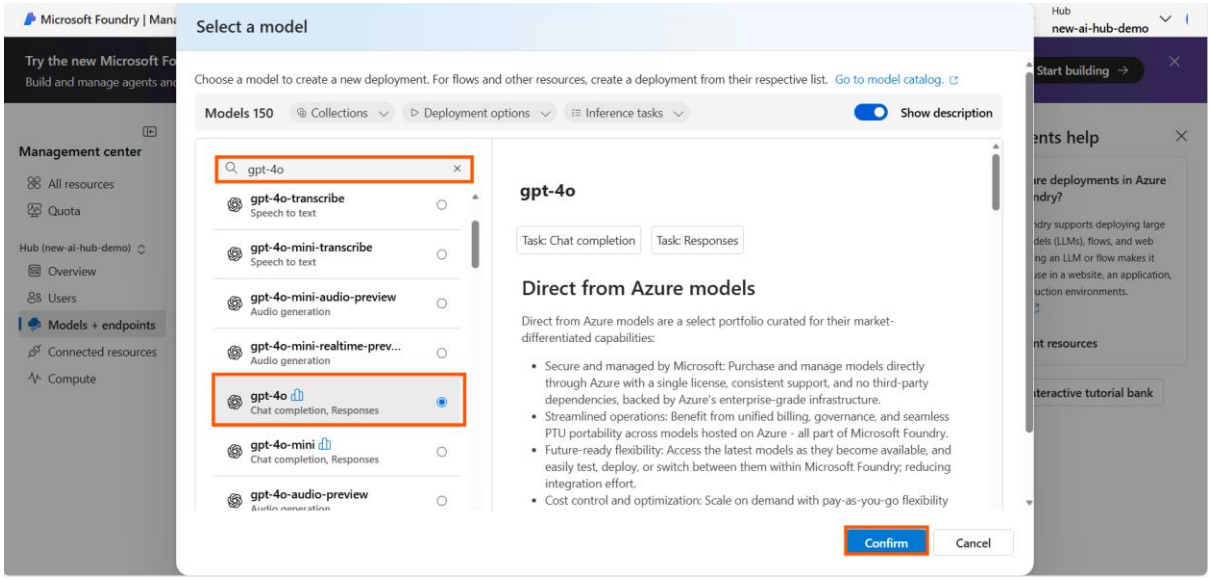
7. Click on **+ New project -> Create**



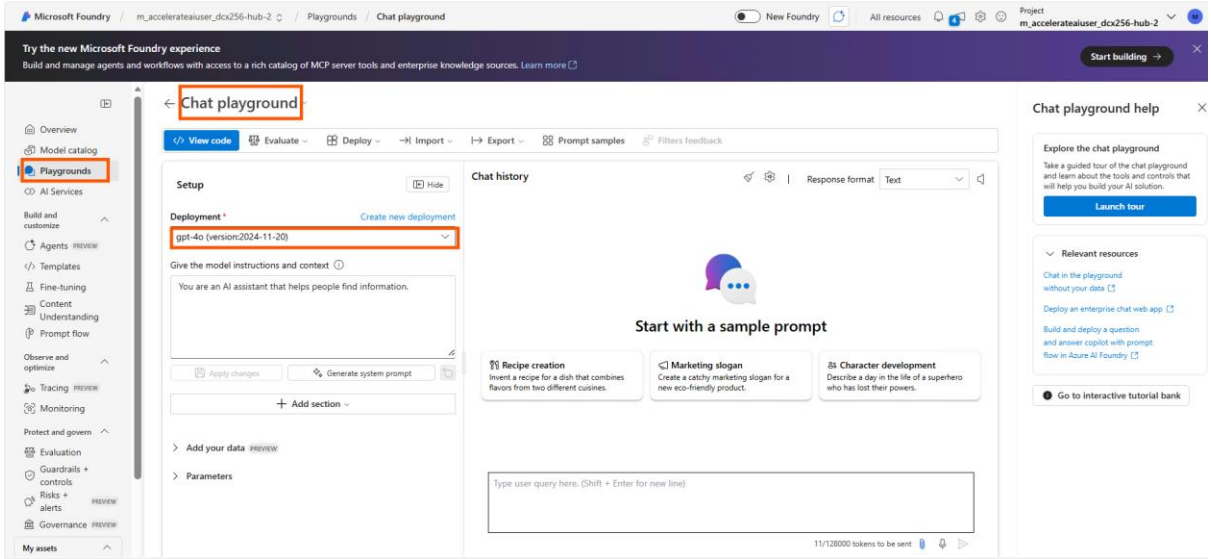
8. Scroll down in the left pane and click on **Models + endpoints -> + Deploy model -> Deploy base model**



9. Search **gpt-4o ->** select **gpt-4o** -> **Confirm -> Connect and deploy.** You might need to change the location to East US/France Central/Sweden Central.

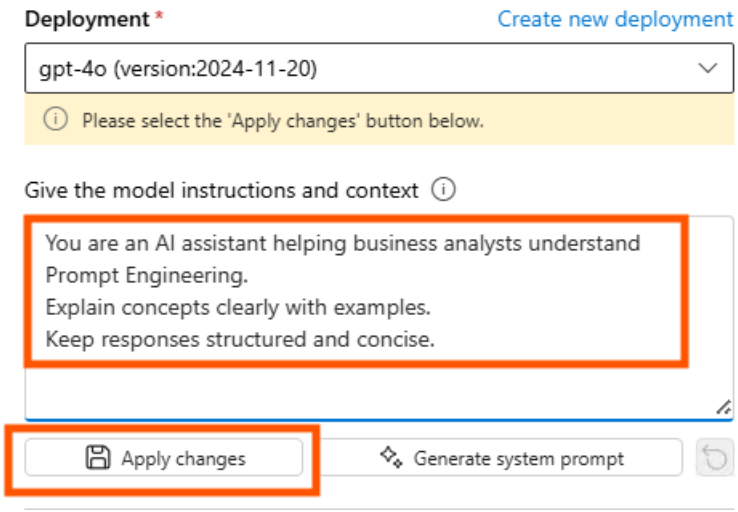


10. In the left pane, click **Playgrounds** → select **Chat playground**. In **Setup**, confirm your **Deployment** is set to **gpt-4o** (the one you deployed earlier).

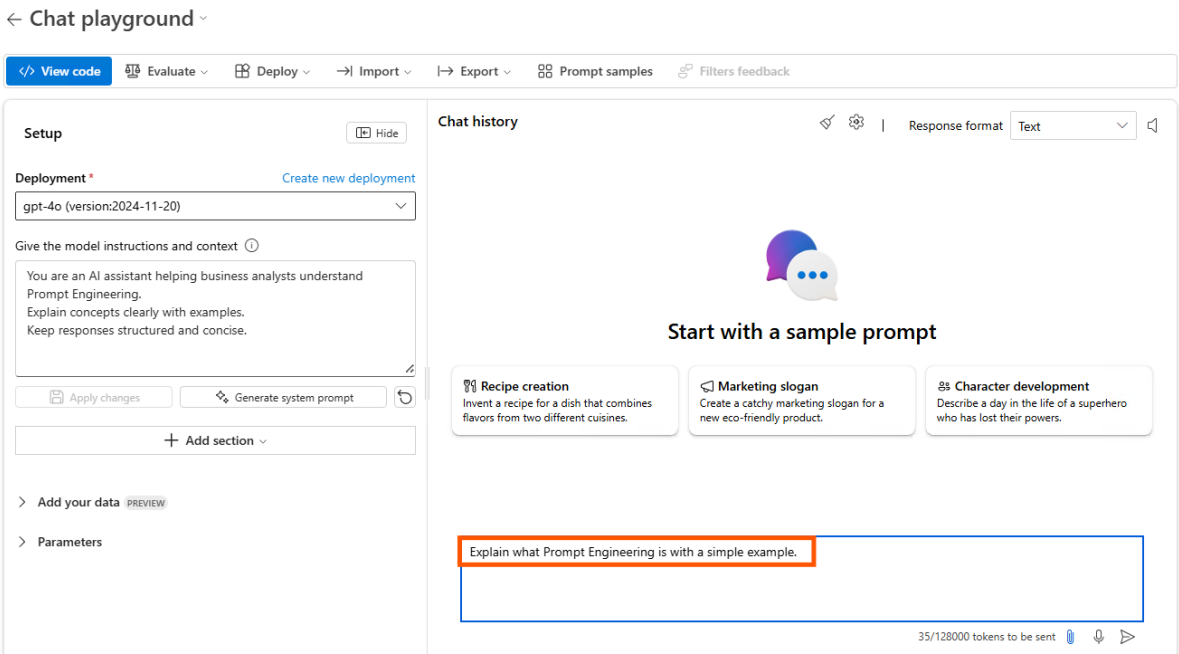


11. In **Give the model instructions and context**, enter your system instructions (define role, tone, rules, etc.), then click **Apply changes > Continue**.

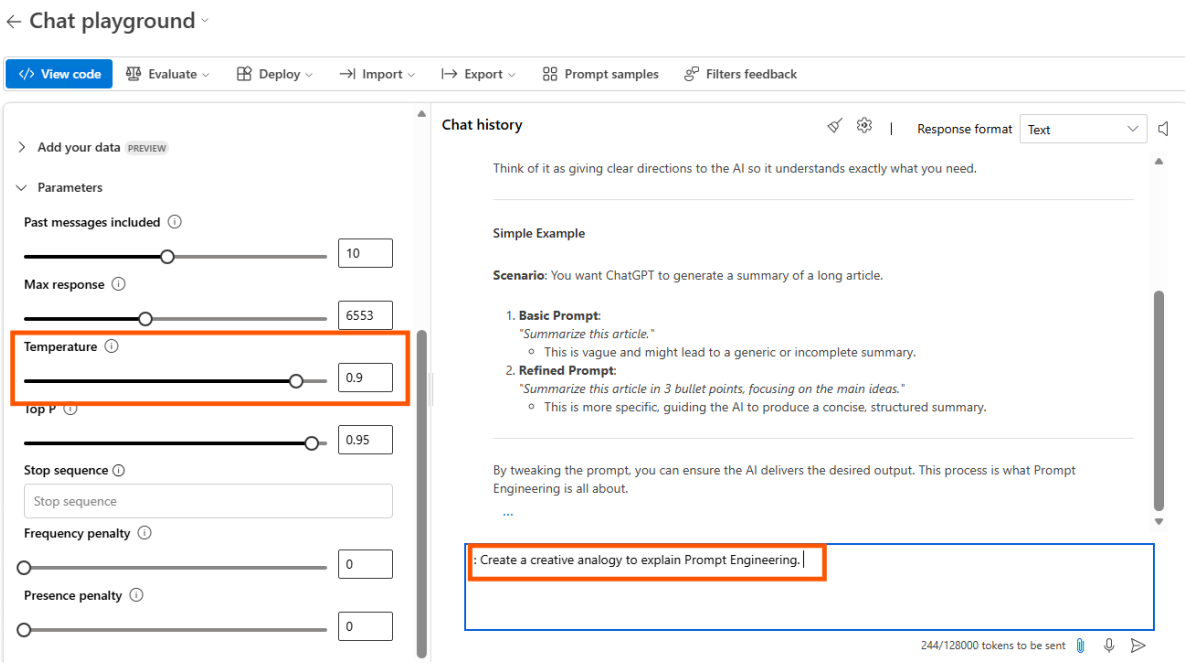
```
You are an AI assistant helping business analysts
understand Prompt Engineering.
Explain concepts clearly with examples.
Keep responses structured and concise.
```



12. In the message box at the bottom (**Type user query here**), enter your prompt and press **Enter** to generate a response.
    `Explain what Prompt Engineering is with a simple example.`



13. Expand **Parameters** (left panel) to adjust settings like creativity (temperature) or response length, then test your prompt again to see the difference.
    Set **Temperature = 0.2** → More deterministic, structured output.
    Set **Temperature = 0.9** → More creative, varied output.
    Re-run this prompt after changing temperature: `Create a creative analogy to explain Prompt Engineering.`



Lab Summary

In this lab, you deployed and interacted with a Large Language Model using Azure AI Foundry's Chat Playground. You configured system instructions, tested prompts, and experimented with temperature settings to observe how model behaviour changes.

You learned how to:

- Control response structure using system instructions
- Experiment with prompt design
- Tune creativity and determinism through parameter adjustments
- Validate outputs before moving toward production scenarios

This foundational skill is critical for designing reliable, controlled, and high-quality AI powered applications.

Congratulations, you have completed the Lab. Please close the current document and move to the next one.